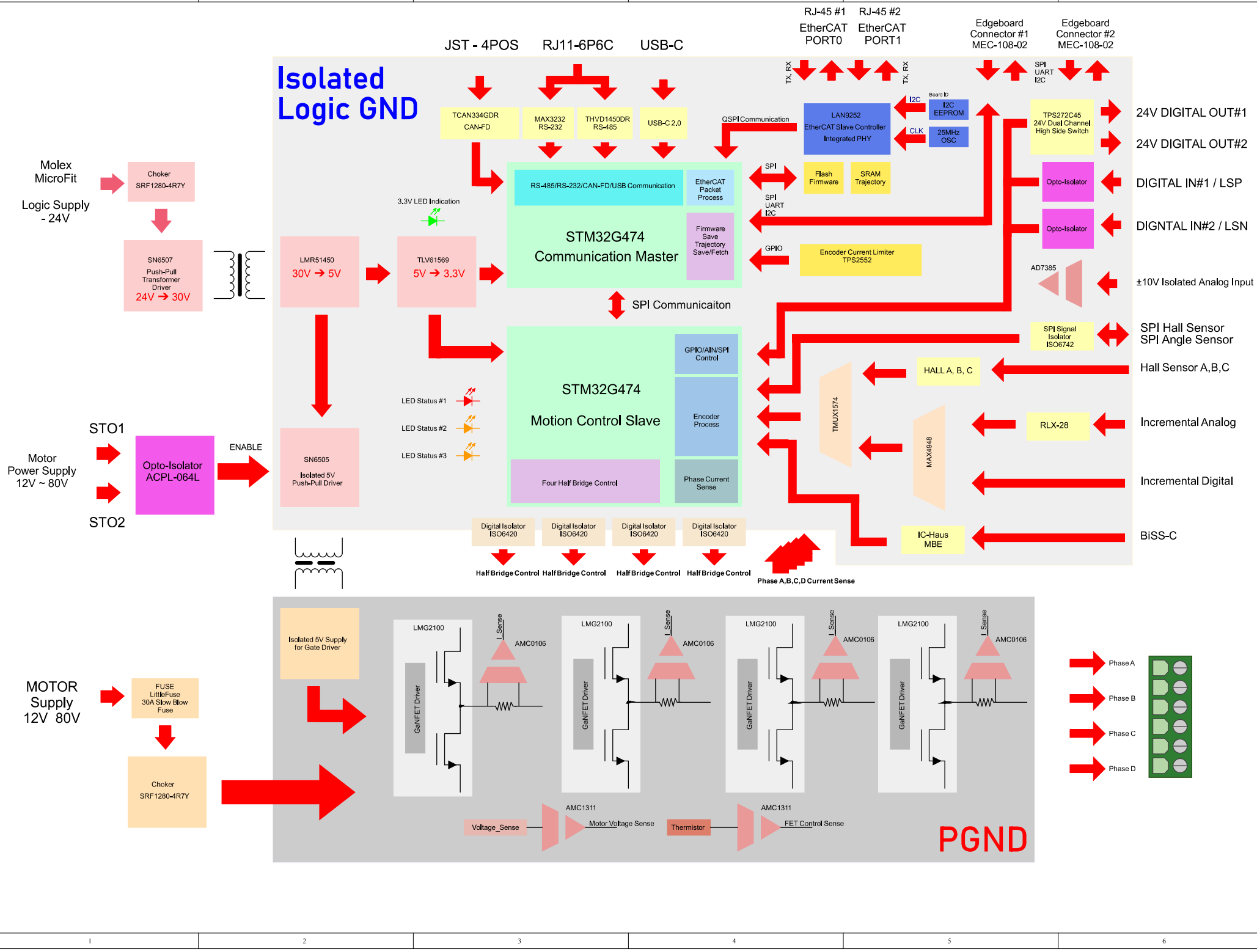
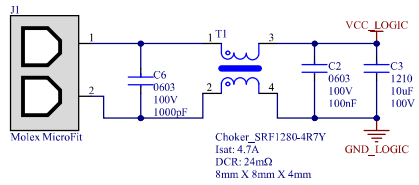
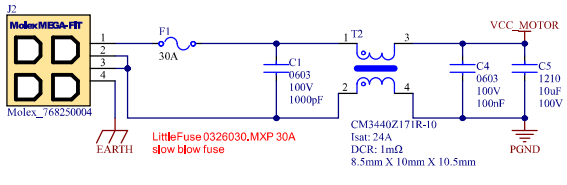

RJ-45 #1
EtherCAT
PORT0
RJ-45 #2
EtherCAT
PORT1
Edgeboard
Connector #1
MEC-108-02
Edgeboard
Connector #2
MEC-108-02
JST - 4POS
RJ11-6P6C
USB-C
Isolated
Logic GND
TCAN334GDR
CAN-FD
MAX3232
RS-232
THVD1450DR
RS-485
USB-C 2.0
QSPI Communication
LAN9252
EtherCAT Slave Controller
Integrated PHY
I2C
I2C
EEPROM
CLK
25MHz
OSC
Board ID
TX, RX
TX, RX
SPI
UART
I2C
TPS272C45
24V Dual Channel
High Side Switch
24V DIGITAL OUT#1
24V DIGITAL OUT#2
Molex
MicroFit
Logic Supply
- 24V
Choker
SRF1280-4R7Y
SN6507
Push-Pull
Transformer
Driver
24V → 30V
LMR51450
30V → 5V
TLV61569
5V → 3.3V
3.3V LED Indication
RS-485/RS-232/CAN-FD/USB Communication
EtherCAT
Packet
Process
STM32G474
Communication Master
Firmware
Save
Trajectory
Save/Fetch
SPI
Flash
Firmware
SRAM
Trajectory
SPI
UART
I2C
GPIO
Encoder Current Limiter
TPS2552
Opto-Isolator
DIGITAL IN#1 / LSP
Opto-Isolator
DIGNTAL IN#2 / LSN
AD7385
±10V Isolated Analog Input
SPI Communicaiton
STM32G474
Motion Control Slave
GPIO/AIN/SPI
Control
Encoder
Process
Four Half Bridge Control
Phase Current
Sense
SPI Signal
Isolator
ISO6742
SPI Hall Sensor
SPI Angle Sensor
HALL A, B, C
Hall Sensor A,B,C
TMUX1574
MAX4948
RLX-28
Incremental Analog
Incremental Digital
IC-Haus
MBE
BiSS-C
LED Status #1
LED Status #2
LED Status #3
STO1
STO2
Motor
Power Supply
12V ~ 80V
Opto-Isolator
ACPL-064L
ENABLE
SN6505
Isolated 5V
Push-Pull Driver
Digital Isolator
ISO6420
Digital Isolator
ISO6420
Digital Isolator
ISO6420
Digital Isolator
ISO6420
Half Bridge Control
Half Bridge Control
Half Bridge Control
Half Bridge Control
Phase A,B,C,D Current Sense
MOTOR
Supply
12V 80V
FUSE
LittleFuse
30A Slow Blow
Fuse
Choker
SRF1280-4R7Y
Isolated 5V Supply
for Gate Driver
LMG2100
GaNFET Driver
I Sense
AMC0106
LMG2100
GaNFET Driver
I Sense
AMC0106
LMG2100
GaNFET Driver
I Sense
AMC0106
LMG2100
GaNFET Driver
I Sense
AMC0106
AMC1311
Voltage_Sense
Motor Voltage Sense
AMC1311
Thermistor
FET Control Sense
PGND
Phase A
Phase B
Phase C
Phase D

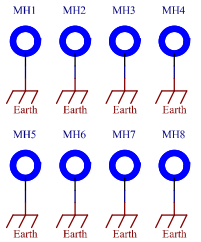
Logic Power Connector



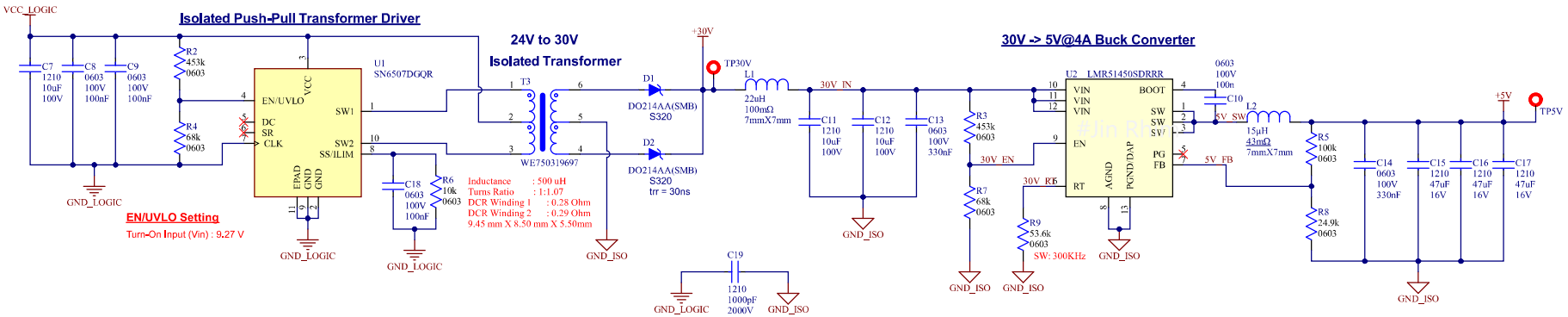
Motor Power Connector



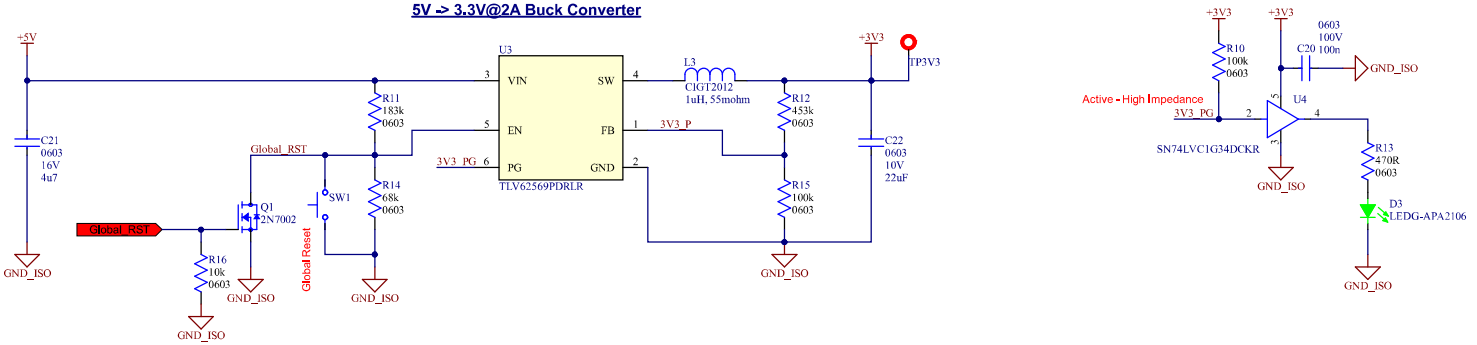
M.3 Mounting Hole



Isolated Logic Power Supply

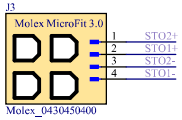


3.3V Logic Power Supply

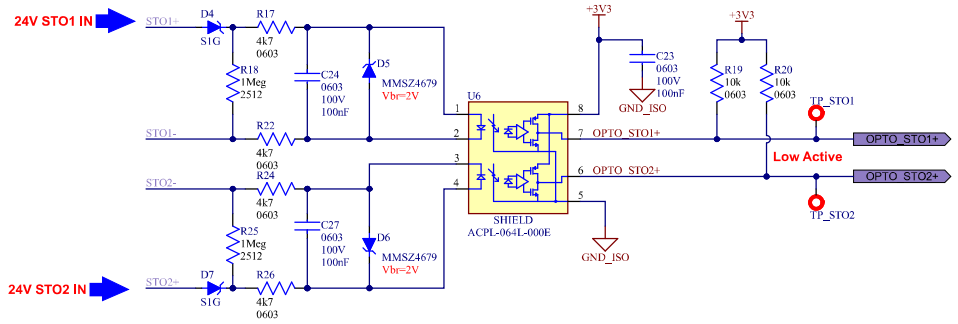


Title	Power Connector #1		
Size: A	Number	Revision	
Date: 03/06/2026		Sheet of	
File: D:\Altium\...\Power.SchDoc		Drawn By: Jin Rhim	

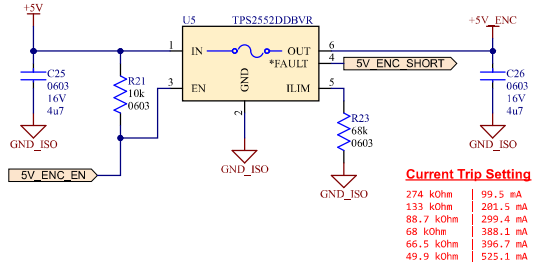
24V STO1, STO2 Connector



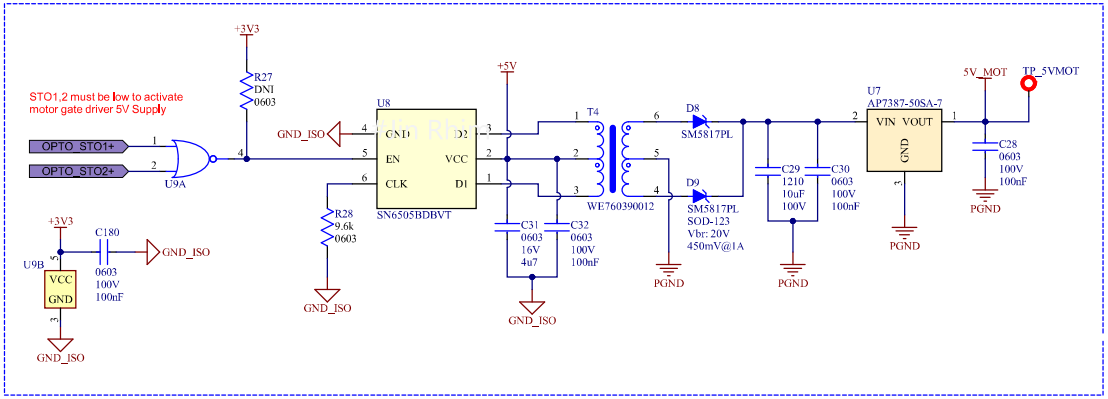
STO1,2 Opto-Isolator



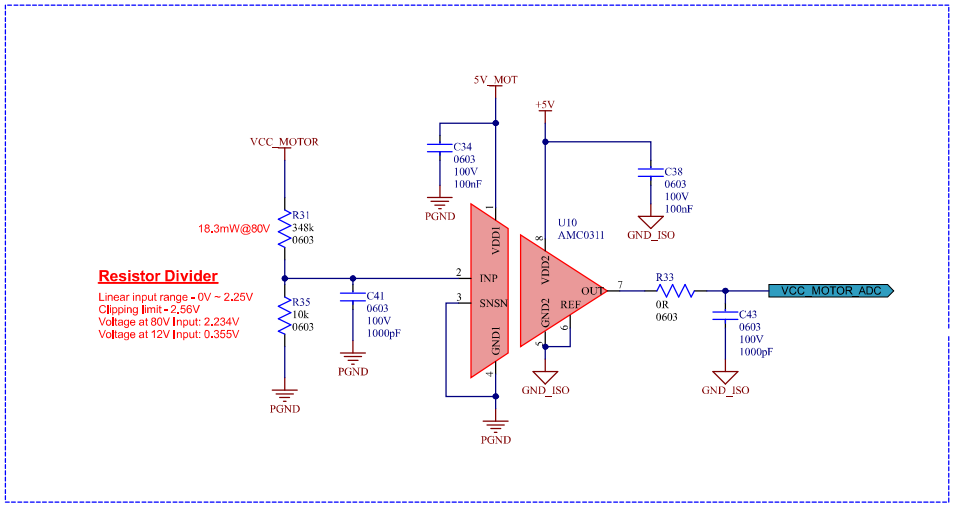
Encoder 5V Supply - Current Limiter



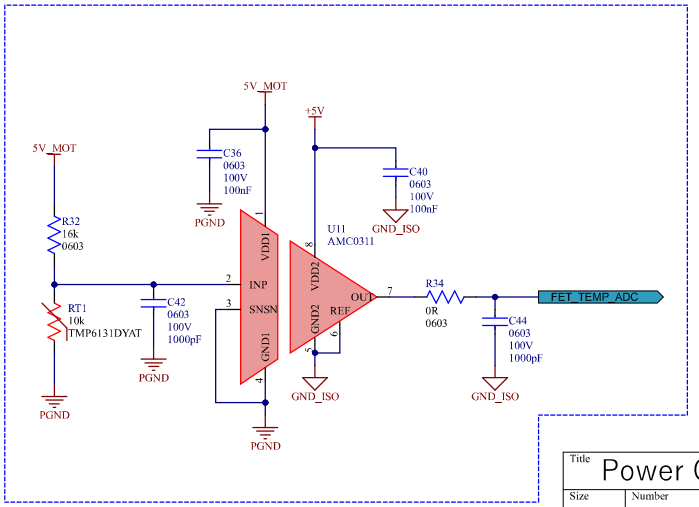
Non-Isolated 5V supply - OPAMP, Gate Driver, Digital Isolator



Isolated Motor Voltage Detection

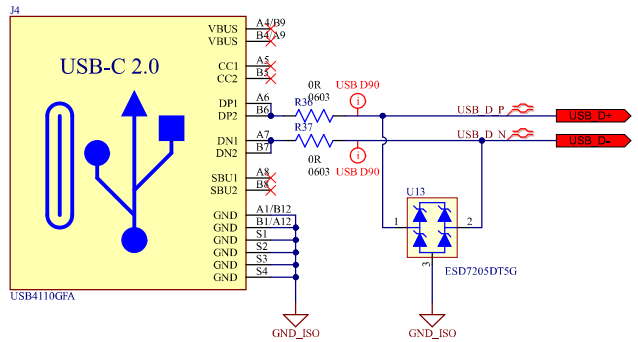


Isolated GaNFET Temperature Sense

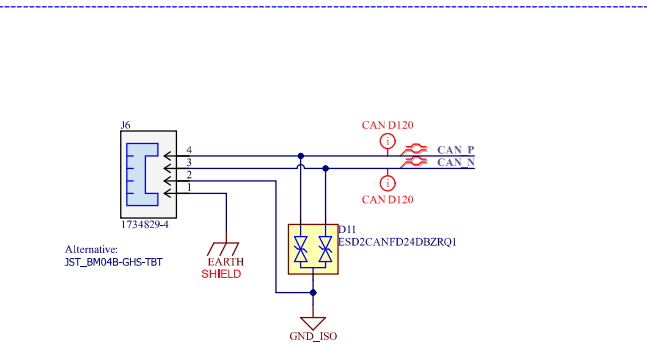


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Size: A	Number	Revision
Date: 03/06/2026		Sheet of
File: D:\Altium\...\Power2.SchDoc		Drawn By: Jin Rhim

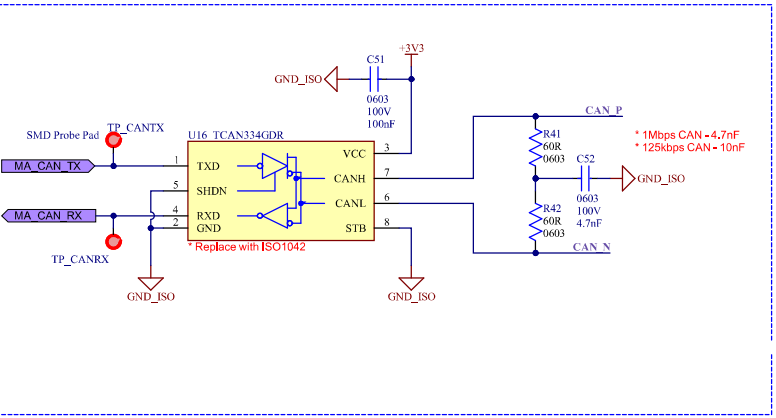
USB C Connector



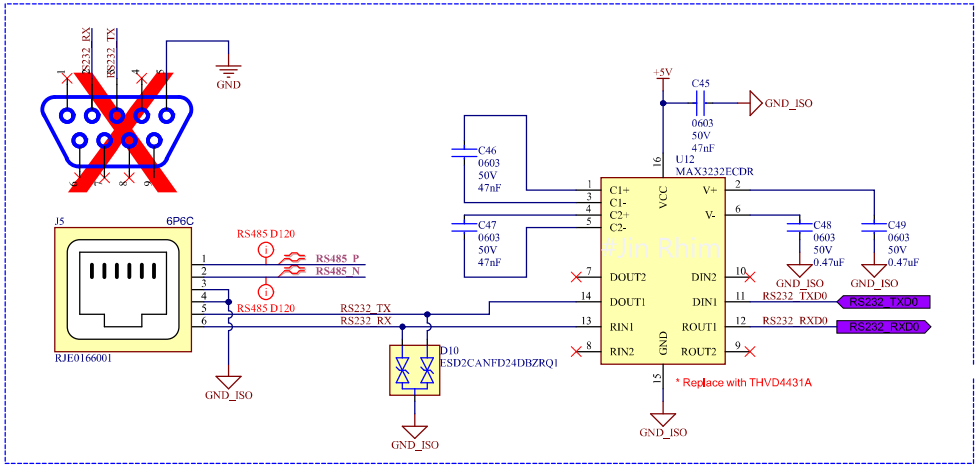
FD-CAN Connector



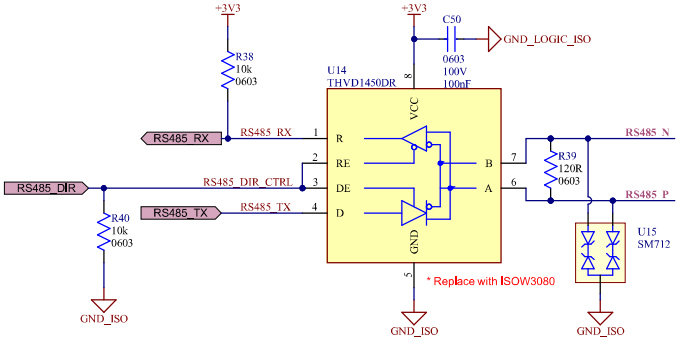
CAN Transceiver



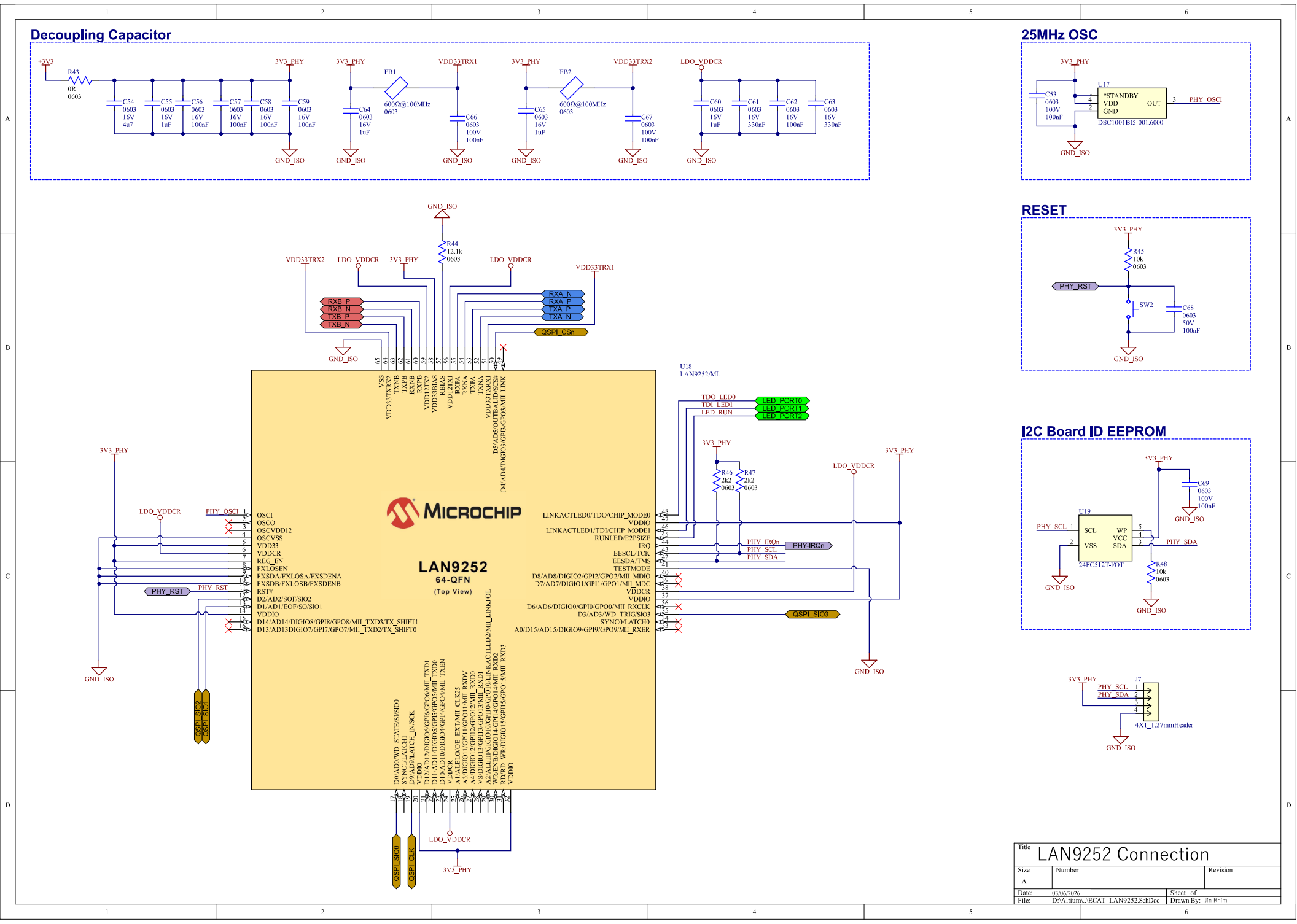
6P6C RJ-11 RS-458/RS-232 Connector



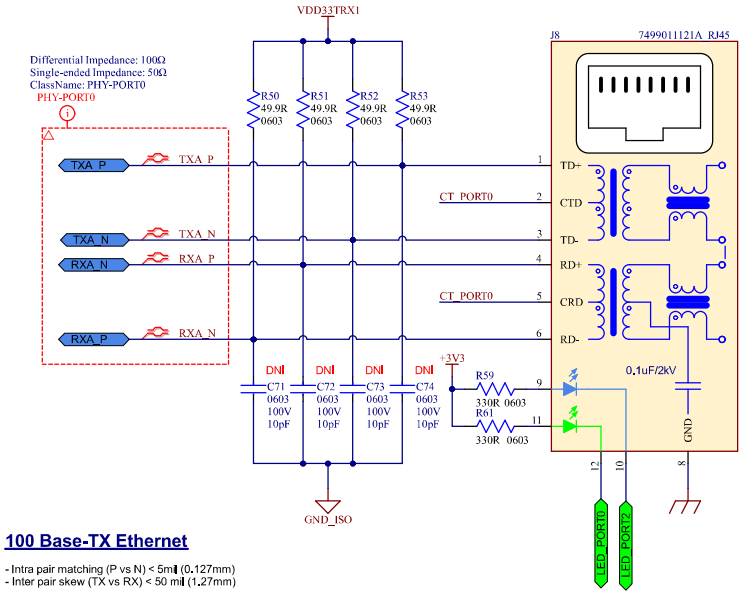
RS-485 Transceiver



Title	Communication Connector		
Size	Number		Revision
A			
Date:	03/06/2026	Sheet of	
File:	D:\Altium\...\Communication.SchDoc	Drawn By:	Jin Rhim

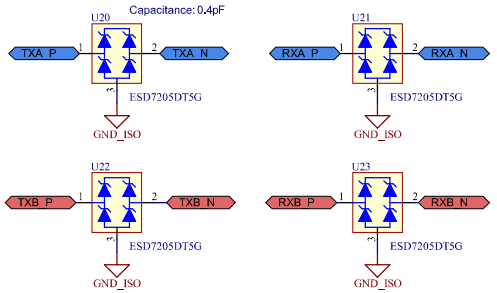

Decoupling Capacitor
+3V3
R43
0R
0603
3V3_PHY
C54 0603 16V 4u7
C55 0603 16V 1uF
C56 0603 16V 100nF
C57 0603 16V 100nF
C58 0603 16V 100nF
C59 0603 16V 100nF
GND_ISO
FB1
600Ω@100MHz
0603
C64 0603 16V 1uF
C66 0603 100V 100nF
VDD33TRX1
FB2
C65 0603 16V 1uF
C67 0603 100V 100nF
VDD33TRX2
LDO_VDDCR
C60 0603 16V 1uF
C61 0603 16V 330nF
C62 0603 16V 100nF
C63 0603 16V 330nF
25MHz OSC
U17
C53 0603 100V 100nF
*STANDBY
VDD
GND
OUT
PHY_OSCI
DSC1001BI5-001.6000
RESET
R45 10k 0603
PHY_RST
SW2
C68 0603 50V 100nF
R44 12.1k 0603
RXB_P
RXB_N
TXB_P
TXB_N
RXA_N
RXA_P
TXA_P
TXA_N
QSPI_CSn
U18
LAN9252/ML
MICROCHIP
LAN9252
64-QFN
(Top View)
OSCI
OSCO
OSCVDD12
OSCVSS
VDD33
VDDCR
REG_EN
FXLOSEN
FXSDA/FXLOSA/FXSDENA
FXSDB/FXLOSB/FXSDENB
RST#
D2/AD2/SOF/SIO2
D1/AD1/EOF/SO/SIO1
VDDIO
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3/TX_SHIFT1
D13/AD13DIGIO7/GPI7/GPO7/MII_TXD2/TX_SHIFT0
LINKACTLED0/TDO/CHIP_MODE0
LINKACTLED1/TDI/CHIP_MODE1
RUNLED/E2PSIZE
IRQ
EESCL/TCK
EESDA/TMS
TESTMODE
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
VDDCR
VDDIO
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D3/AD3/WD_TRIG/SIO3
SYNC0/LATCH0
A0/D15/AD15/DIGIO9/GPI9/GPO9/MII_RXER
TDO_LED0
TDI_LED1
LED_RUN
LED_PORT0
LED_PORT1
LED_PORT2
R46 2k2 0603
R47 2k2 0603
PHY_IRQn
PHY-IRQn
PHY_SCL
PHY_SDA
QSPI_SIO3
QSPI_SIO2
QSPI_SIO1
QSPI_SIO0
QSPI_CLK
I2C Board ID EEPROM
U19
C69 0603 100V 100nF
SCL
SDA
WP
VCC
VSS
24FC512T-I/OT
R48 10k 0603
J7
4X1_1.27mmHeader
GND_ISO
LDO_VDDCR
3V3_PHY
Title
LAN9252 Connection
Size
A
Number
Revision
Date: 03/06/2026
File: D:\Altium\...\ECAT_LAN9252.SchDoc
Sheet of
Drawn By: Jin Rhim

ECAT IN

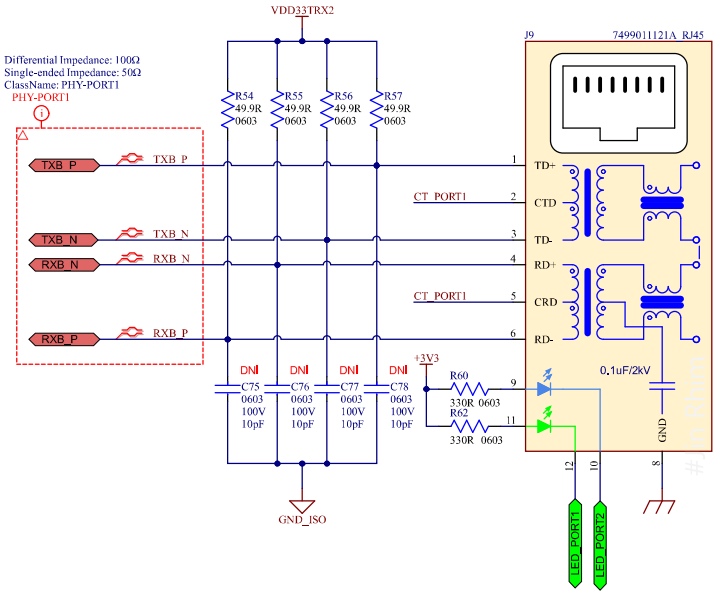


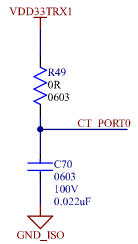
100 Base-TX Ethernet

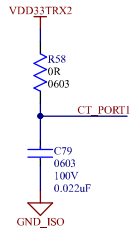
- Intra pair matching (P vs N) < 5mil (0.127mm)
- Inter pair skew (TX vs RX) < 50 mil (1.27mm)



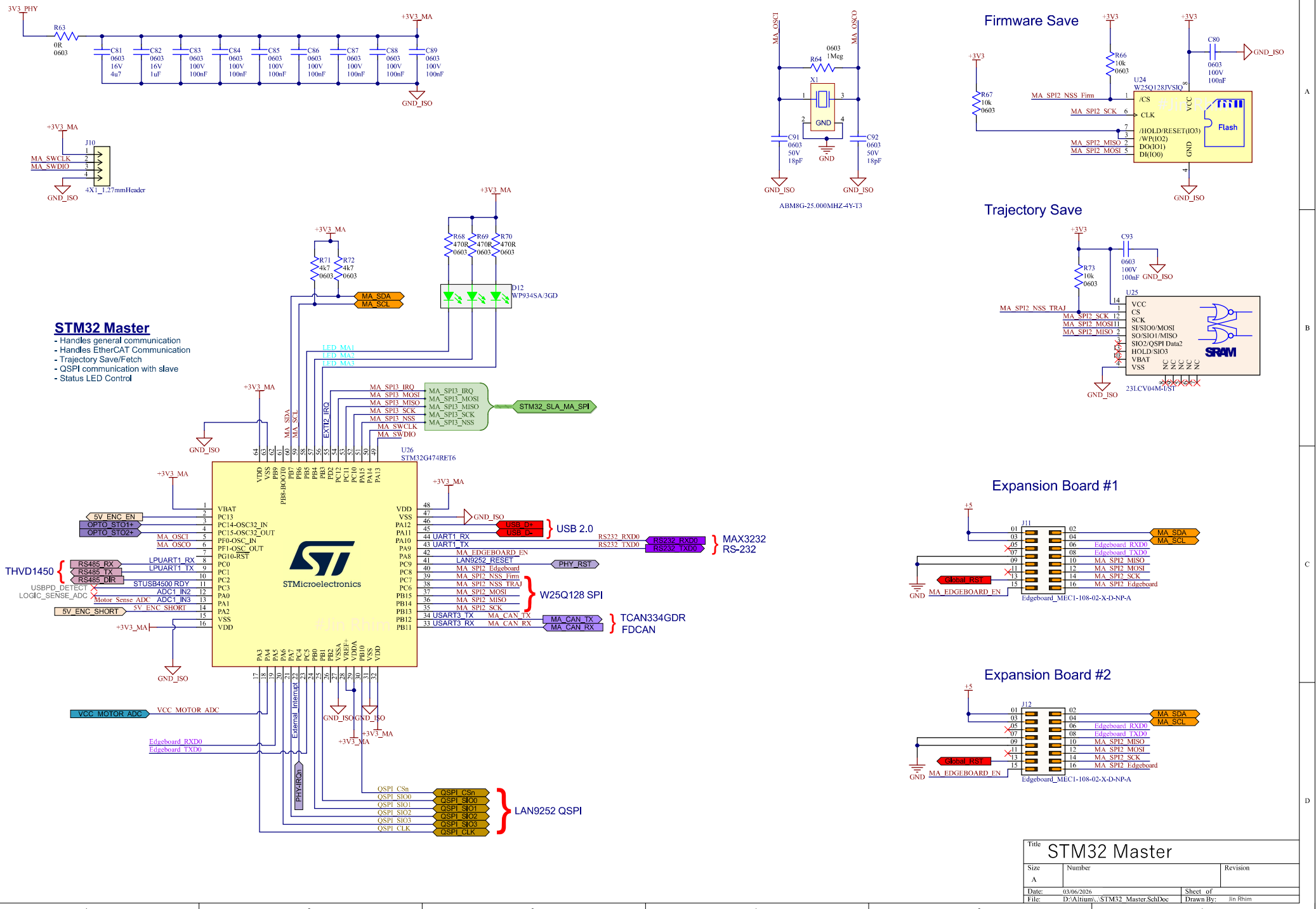
ECAT OUT

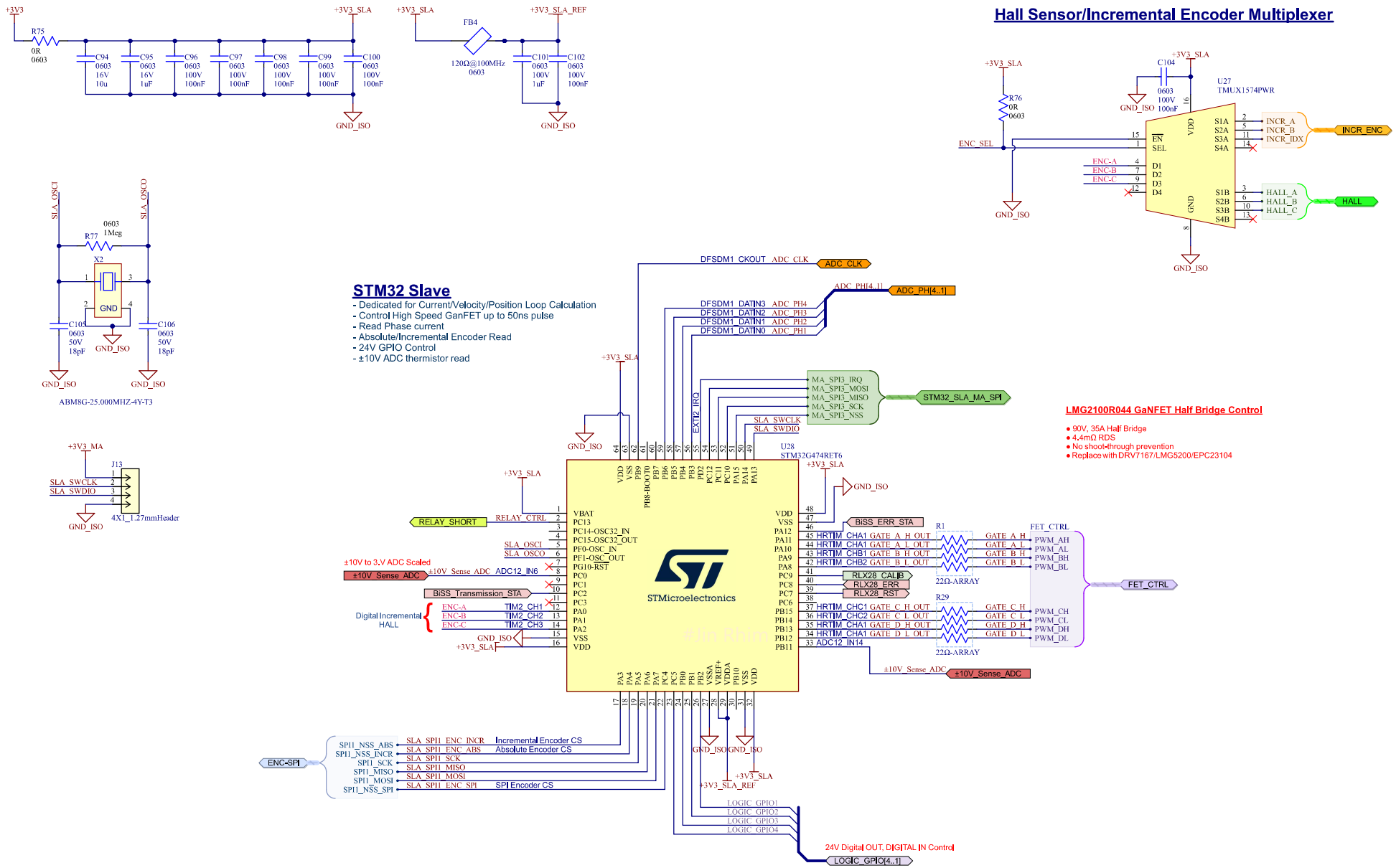


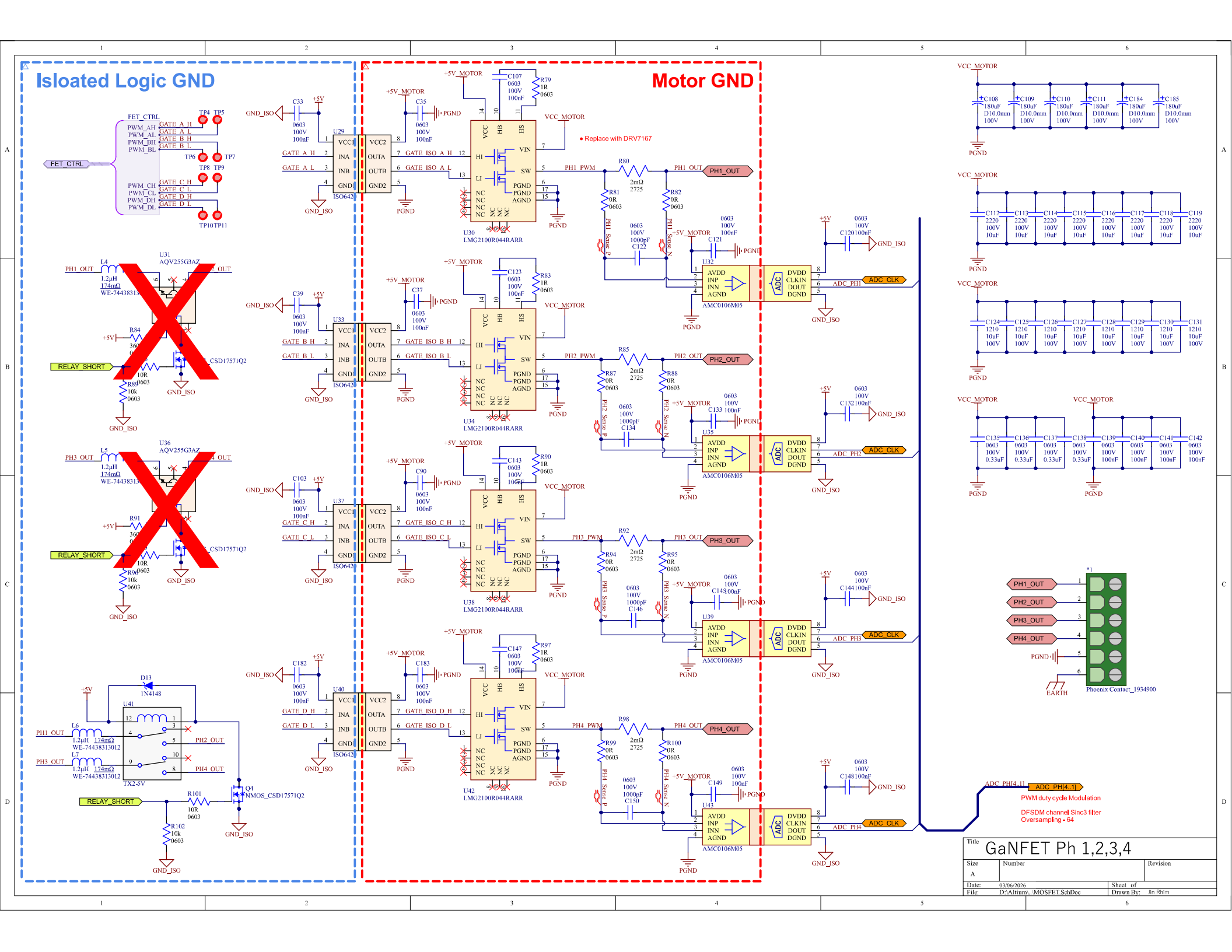


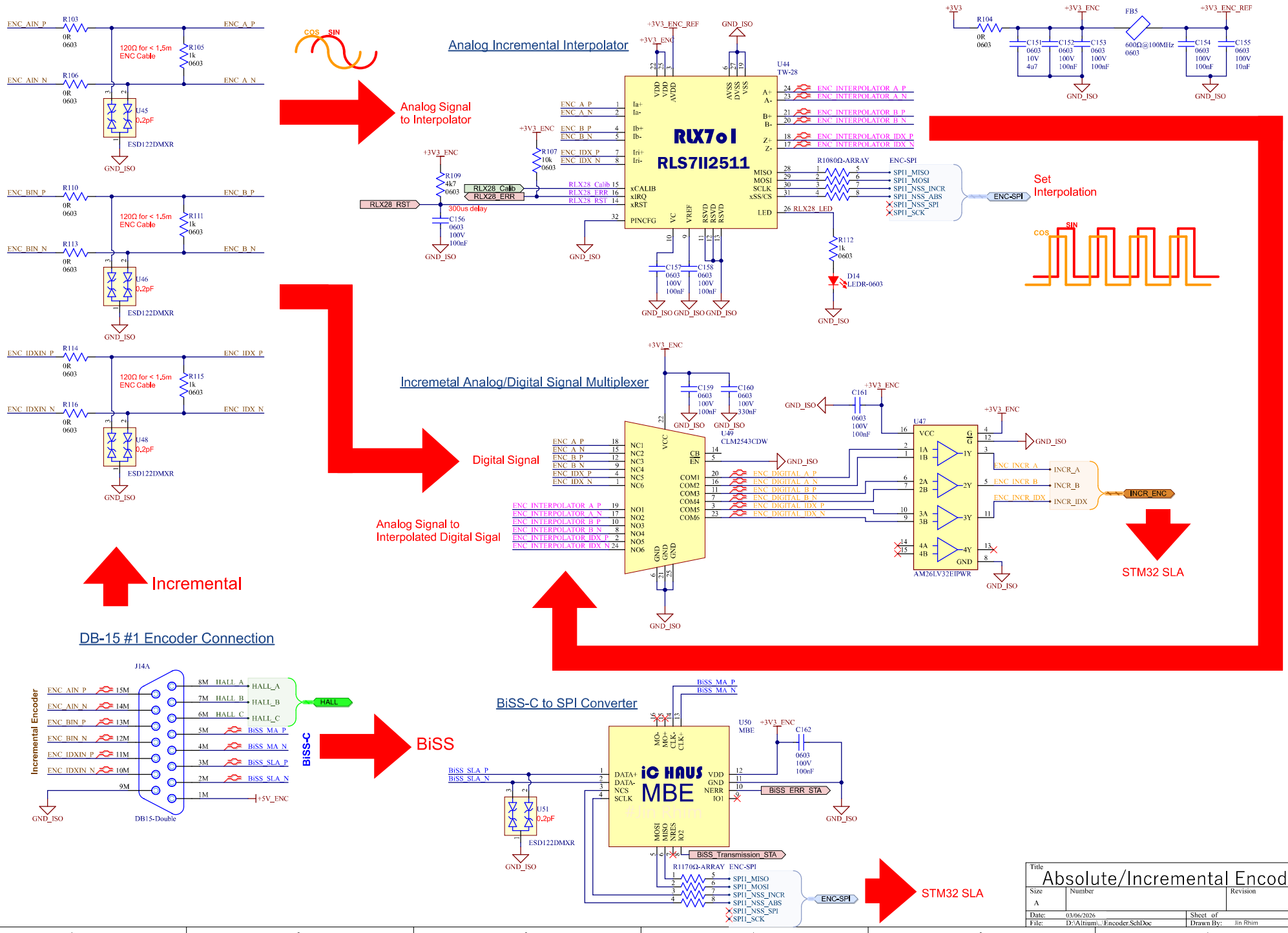


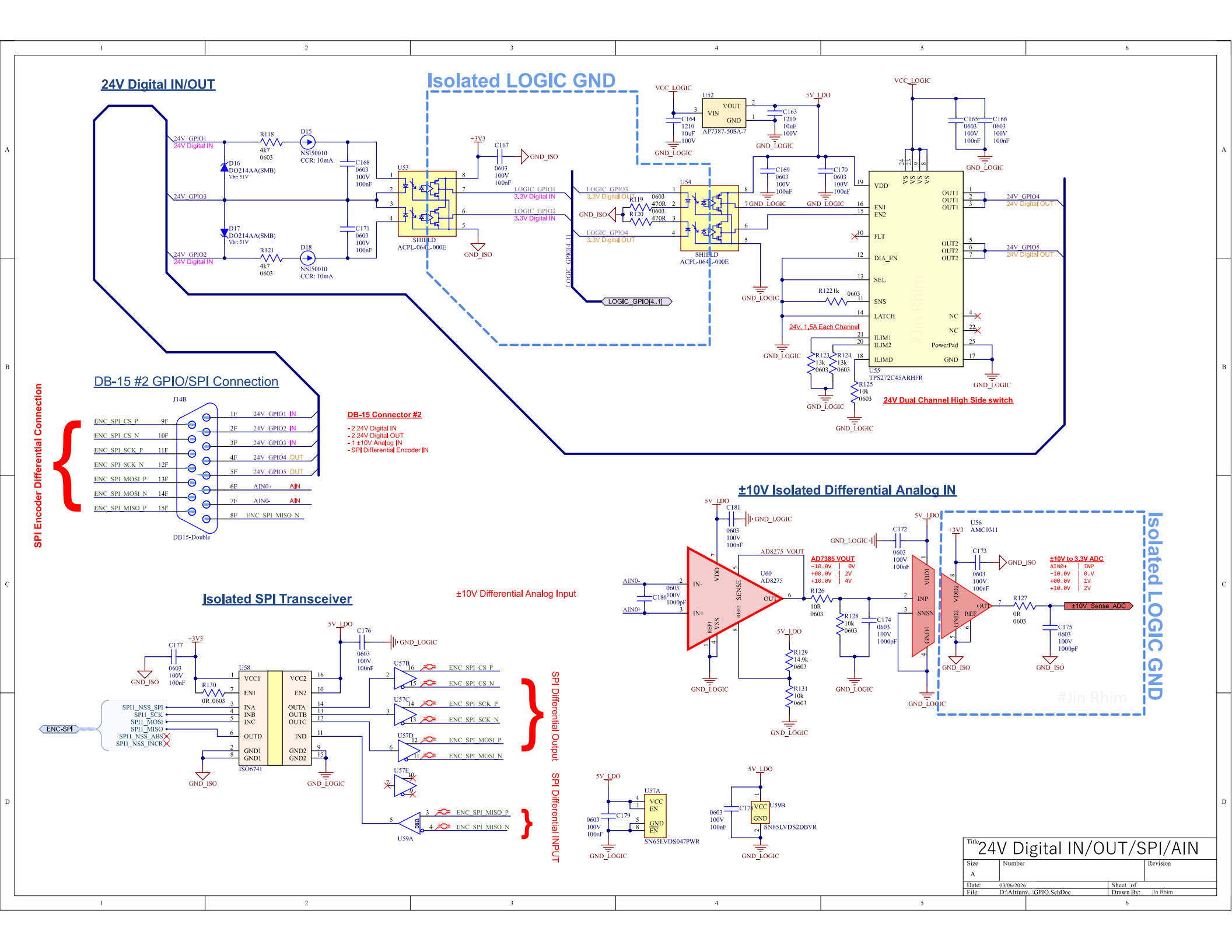
Title: RJ45 PHY Connection		
Size: A	Number	Revision
Date: 03/06/2026		Sheet of
File: D:\Altium\...\ECAT_RJ45_Conn.SchDoc		Drawn By: Jin Rhim


STM32 Master
- Handles general communication
- Handles EtherCAT Communication
- Trajectory Save/Fetch
- QSPI communication with slave
- Status LED Control
Firmware Save
Trajectory Save
Expansion Board #1
Expansion Board #2
STM32_SLA_MA_SPI
USB 2.0
MAX3232
RS-232
THVD1450
W25Q128 SPI
TCAN334GDR
FDCAN
LAN9252 QSPI
ABM8G-25.000MHZ-4Y-T3
4X1_1.27mmHeader
Edgeboard_MEC1-108-02-X-D-NP-A
STMicroelectronics
Title
STM32 Master
Size
A
Number
Revision
Date:
03/06/2026
File:
D:\Altium\...\STM32_Master.SchDoc
Sheet of
Drawn By:
Jin Rhim


Hall Sensor/Incremental Encoder Multiplexer
STM32 Slave
- Dedicated for Current/Velocity/Position Loop Calculation
- Control High Speed GanFET up to 50ns pulse
- Read Phase current
- Absolute/Incremental Encoder Read
- 24V GPIO Control
- ±10V ADC thermistor read
LMG2100R044 GaNFET Half Bridge Control
• 90V, 35A Half Bridge
• 4.4mΩ RDS
• No shoot-through prevention
• Replace with DRV7167/LMG5200/EPC23104
STMicroelectronics
U28
STM32G474RET6
U27
TMUX1574PWR
ABM8G-25.000MHZ-4Y-T3
4X1_1.27mmHeader
Title STM32 SLAVE
Size A
Number
Revision
Date: 03/06/2026
File: D:\Altium\...\STM32_Slave.SchDoc
Sheet of
Drawn By: Jin Rhim


Isloated Logic GND
Motor GND
FET_CTRL
PWM_AH
PWM_AL
PWM_BH
PWM_BL
PWM_CH
PWM_CL
PWM_DH
PWM_DL
GATE_A_H
GATE_A_L
GATE_B_H
GATE_B_L
GATE_C_H
GATE_C_L
GATE_D_H
GATE_D_L
TP4 TP5 TP6 TP7 TP8 TP9 TP10TP11
U29 U30 U33 U34 U37 U38 U40 U42
ISO6420
LMG2100R044RARR
• Replace with DRV7167
+5V_MOTOR
VCC_MOTOR
PGND
GND_ISO
PH1_PWM
PH2_PWM
PH3_PWM
PH4_PWM
PH1_OUT
PH2_OUT
PH3_OUT
PH4_OUT
PH1_Sense_P
PH1_Sense_N
PH2_Sense_P
PH2_Sense_N
PH3_Sense_P
PH3_Sense_N
PH4_Sense_P
PH4_Sense_N
U31 U36
AQV255G3AZ
L4 L5 L6 L7
1.2µH 174mΩ
WE-74438313012
RELAY_SHORT
NMOS_CSD17571Q2
D13
1N4148
U41
TX2-5V
U32 U35 U39 U43
AMC0106M05
ADC_PH1
ADC_PH2
ADC_PH3
ADC_PH4
ADC_CLK
ADC_PH[4..1]
PWM duty cycle Modulation
DFSDM channel Sinc3 filter
Oversampling - 64
Phoenix Contact_1934900
EARTH
Title
GaNFET Ph 1,2,3,4
Size
A
Number
Revision
Date: 03/06/2026
File: D:\Altium\..\MOSFET.SchDoc
Sheet of
Drawn By: Jin Rhim


ENC_AIN_P
ENC_AIN_N
ENC_BIN_P
ENC_BIN_N
ENC_IDXIN_P
ENC_IDXIN_N
120Ω for < 1.5m ENC Cable
ESD122DMXR
GND_ISO
COS SIN
Analog Signal to Interpolator
Analog Incremental Interpolator
RLX7o1 RLS7II2511
TW-28
+3V3_ENC
+3V3_ENC_REF
GND_ISO
ENC_INTERPOLATOR_A_P
ENC_INTERPOLATOR_A_N
ENC_INTERPOLATOR_B_P
ENC_INTERPOLATOR_B_N
ENC_INTERPOLATOR_IDX_P
ENC_INTERPOLATOR_IDX_N
R1080Ω-ARRAY
ENC-SPI
SPI1_MISO
SPI1_MOSI
SPI1_NSS_INCR
SPI1_NSS_ABS
SPI1_NSS_SPI
SPI1_SCK
RLX28_Calib
RLX28_ERR
RLX28_RST
300us delay
LED
D14
LEDR-0603
Set Interpolation
Incremetal Analog/Digital Signal Multiplexer
Digital Signal
Analog Signal to Interpolated Digital Sigal
U49
CLM2543CDW
U47
AM26LV32EIPWR
INCR_A
INCR_B
INCR_IDX
INCR_ENC
STM32 SLA
Incremental
DB-15 #1 Encoder Connection
Incremental Encoder
HALL_A
HALL_B
HALL_C
BiSS_MA_P
BiSS_MA_N
BiSS_SLA_P
BiSS_SLA_N
+5V_ENC
DB15-Double
HALL
BISS-C
BiSS
BiSS-C to SPI Converter
iC HAUS MBE
BiSS_ERR_STA
BiSS_Transmission_STA
R1170Ω-ARRAY
STM32 SLA
Absolute/Incremental Encoder
Size A
Number
Revision
Date: 03/06/2026
File: D:\Altium\..\Encoder.SchDoc
Sheet of
Drawn By: Jin Rhim

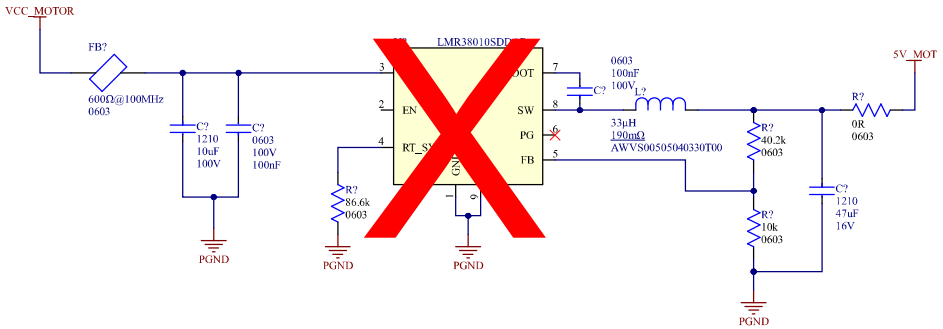


Title	24V Digital IN/OUT/SPI/AIN		
Size	Number		Revision
A			
Date:	03/06/2026	Sheet of	
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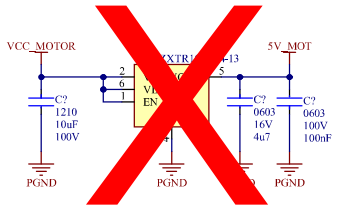
* Maximum 15mm/s
* Minimum 125nF

Title	Piezo Driver		
Size: A	Number:		Revision:
Date: 03/06/2026		Sheet of	
File: D:\Altium\..\Piezo_Driver.SchDoc		Drawn By: Jin Rhim	

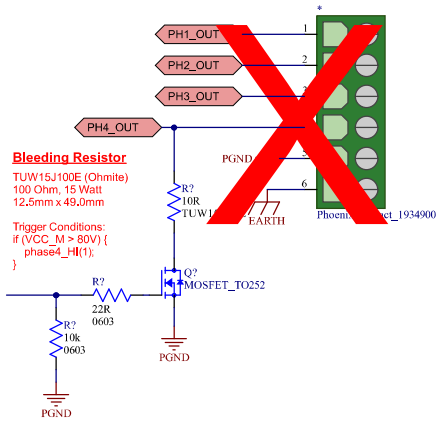
80V ➜ 5V Buck Logic Power Supply



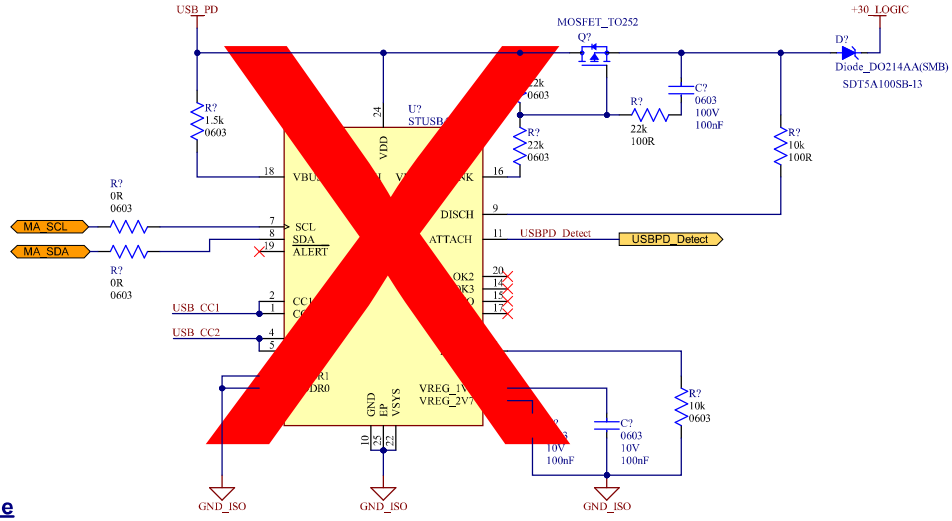
80V ➜ 5V LDO



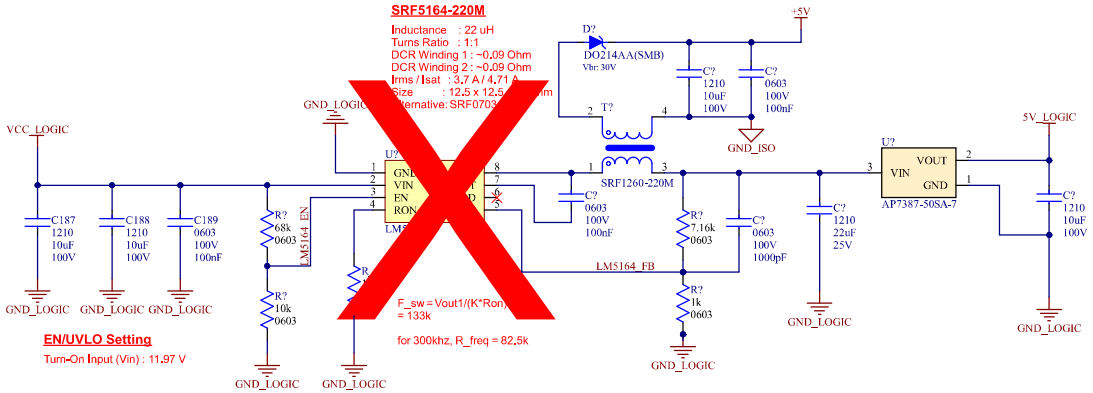
Power Cropping Resistor for Phase 4



20V, 5A USB-C PD



LM5164 Fly-Buck Converter for Dual isolated/non-isolated logic voltage



#Ebimo_Hashina

Title: DO NOT INSTALL		
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Date: 03/06/2026		Sheet of
File: D:\Altium\...\DNI.SchDoc		Drawn By: Jin Rhim